

National Textile University, Faisalabad



Department of Computer Science

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Project name	Smart plant gardening
Course Name:	Embedded and IOT system
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1. Problem Statement and Objectives

1.1 Problem Statement

Plants require proper and timely watering to grow healthy. In traditional plant watering systems, watering is done manually, which depends completely on human availability and attention. Due to busy schedules, forgetfulness, or absence from home, plants often suffer from over-watering or under-watering. Both conditions negatively affect plant health and may lead to plant damage.

Additionally, when a person is outside the city, it becomes impossible to monitor soil conditions or control the water pump remotely. Traditional systems also lack an alert mechanism to inform the user when the soil becomes dry.

Therefore, there is a need for an automated and remotely accessible system that can monitor soil moisture in real time, notify the user when watering is required, and allow remote control of the water pump.

1.2 Objectives

The main objectives of this project are:

- To design an IoT-based smart plant gardening system
- To monitor soil moisture level continuously
- To send an email alert when soil becomes dry
- To allow remote ON/OFF control of water pump using Blynk mobile application
- To automatically turn OFF the pump when soil becomes wet
- To reduce water wastage
- To provide plant care even when the user is out of city.

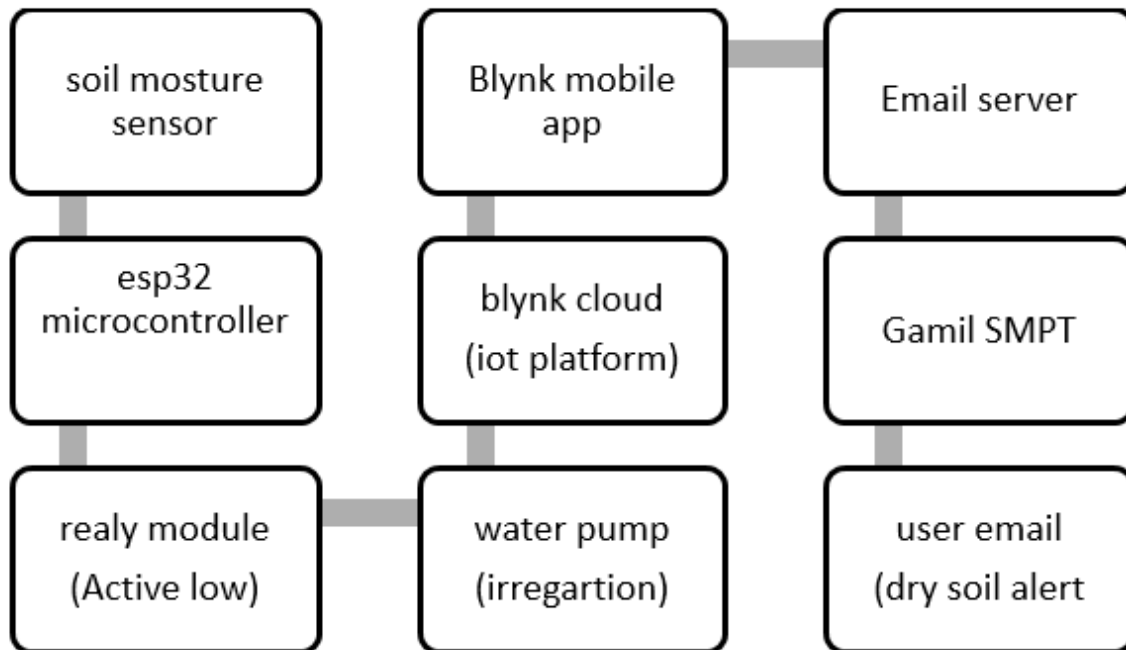
2. System Architecture / Block Diagram

2.1 System Architecture Description

The Smart Plant Gardening System is based on the Internet of Things (IoT) concept. The system uses an ESP32 microcontroller as the main processing unit. A soil moisture sensor is connected to the ESP32 to measure soil condition. The ESP32 connects to the internet using WiFi and communicates with the Blynk cloud platform.

When the soil becomes dry, the ESP32 sends an email alert to the user through an SMTP server. The user can then turn ON the water pump remotely using the Blynk mobile application. A relay module is used to control the water pump. When the soil becomes wet, the system automatically turns OFF the pump.

2.2 Block Diagram Explanation



The block diagram consists of the following blocks:

- Soil Moisture Sensor
- ESP32 Microcontroller
- WiFi Network
- Blynk Cloud Server
- Mobile Application (Blynk App)
- Relay Module

- Water Pump
- Data flows from the sensor to the ESP32, then to the Blynk cloud, and finally to the user interface.

3. Hardware and Software Description

3.1 Hardware Description

3.1.1 ESP32 Microcontroller

ESP32 is a low-cost and powerful microcontroller with built-in WiFi. It acts as the brain of the system and controls all operations including sensor reading, internet communication, and relay control.

Features:

- Built-in WiFi
- Multiple GPIO and ADC pins
- Low power consumption

3.1.2 Soil Moisture Sensor

The soil moisture sensor measures the amount of water present in the soil. It provides an analog output that changes according to soil condition.

- Dry soil → High analog value
- Wet soil → Low analog value

3.1.3 Relay Module

The relay module is used to control the water pump. It allows the ESP32 to safely switch high-voltage devices using low-voltage signals.

3.1.4 Water Pump

The water pump supplies water to the plant when activated by the relay module.

3.2 Software Description

3.2.1 Arduino IDE

Arduino IDE is used to write, compile, and upload code to the ESP32 microcontroller.

3.2.2 Blynk IoT Platform

Blynk is an IoT platform that provides a mobile application for monitoring and controlling hardware devices remotely.

Widgets Used:

- Gauge (Soil Moisture Value)
- Button (Pump Control)
- LED Indicator (Soil Status)

3.2.3 ESP Mail Client Library

This library is used to send email alerts using Gmail SMTP service when the soil becomes dry.

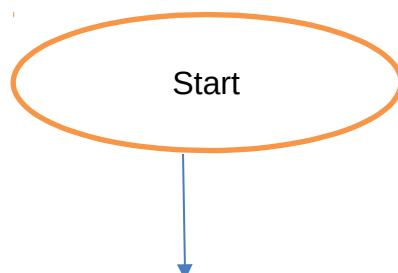
4. Methodology and Flowchart

4.1 Methodology

The system follows the steps below:

1. ESP32 connects to WiFi network
2. Soil moisture sensor reads soil condition
3. ESP32 processes sensor data
4. If soil is dry:
 - Email alert is sent to user
 - Soil status LED turns ON in Blynk
5. User turns ON water pump through Blynk app
6. Relay activates and pump starts watering
7. Soil moisture increases
8. When soil becomes wet:
 - Pump turns OFF automatically
 - System resets for next cycle

4.2 Flow chart



Power on esp32
(cable/battery)

Initialize pins
Soil ,relay

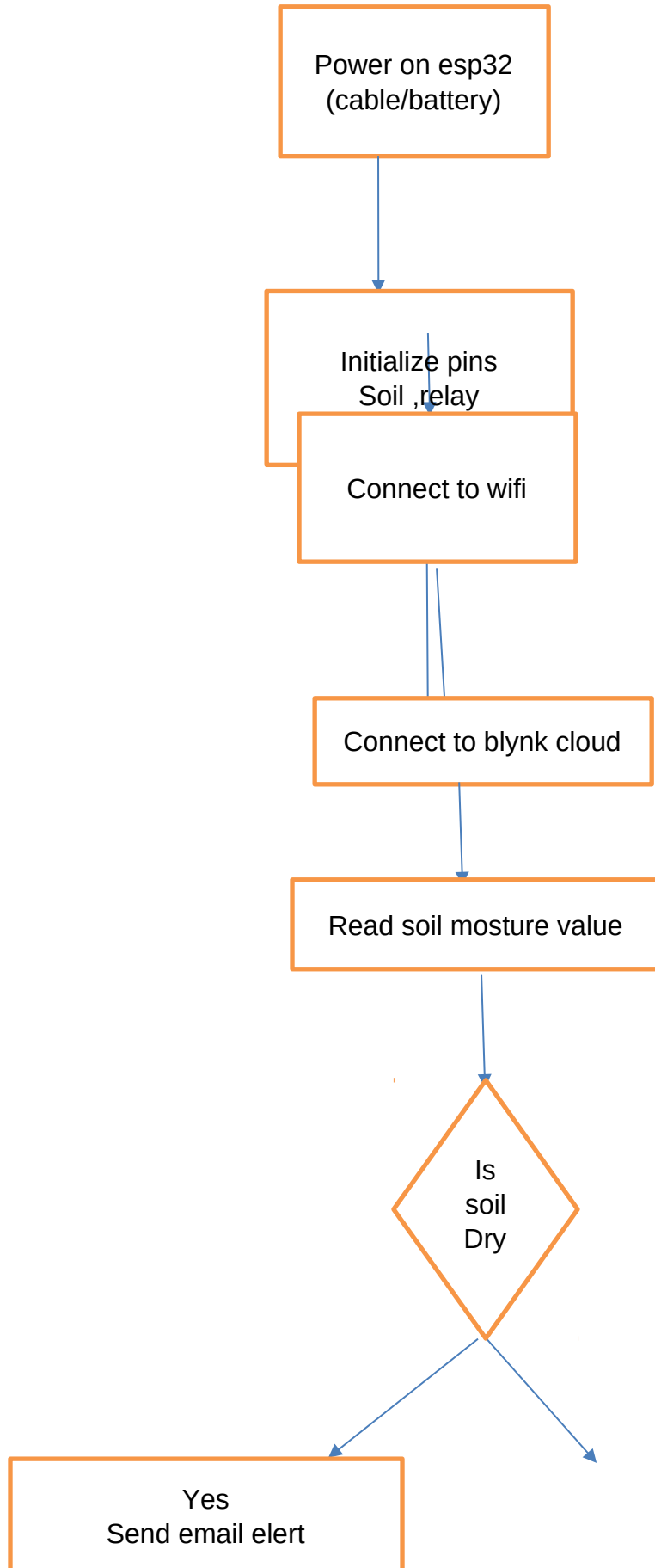
Connect to wifi

Connect to blynk cloud

Read soil moisture value

Is
soil
Dry

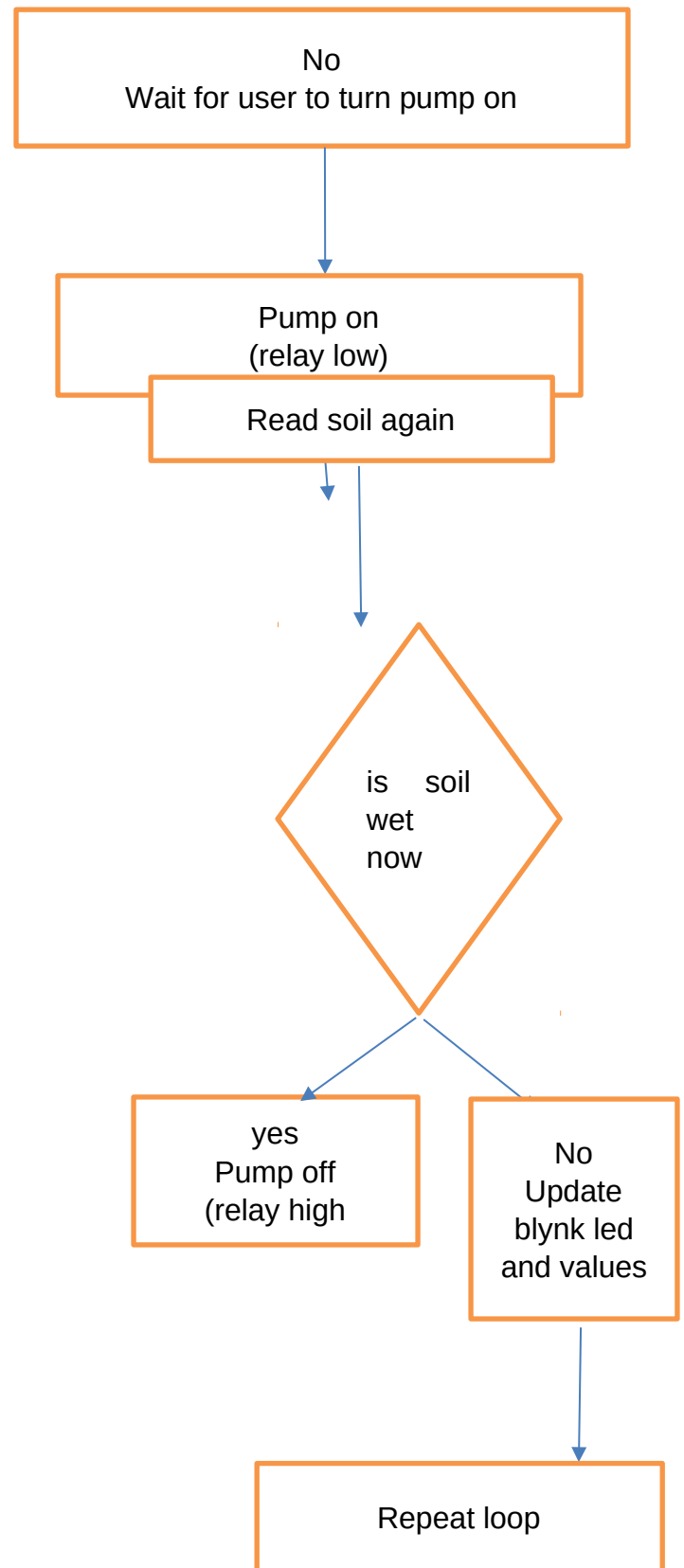
Yes
Send email elert



yes

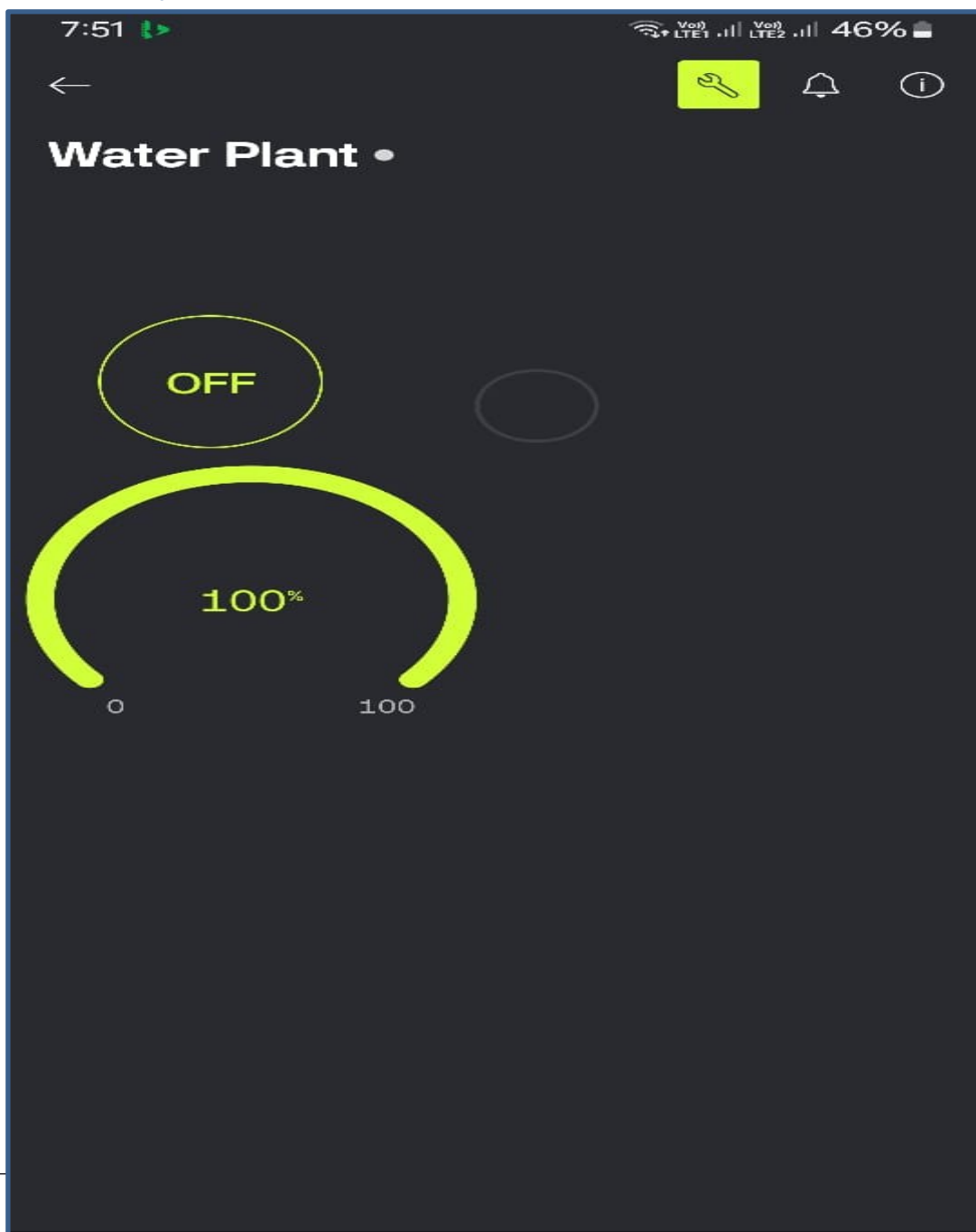
The flowchart shows:

- Start
- Initialization
- Soil moisture reading
- Dry/Wet condition check
- Email alert
- User control
- Pump ON/OFF
- Loop repetition



5. Screenshot of Output and Dashboard

5.1 Blynk Dashboard



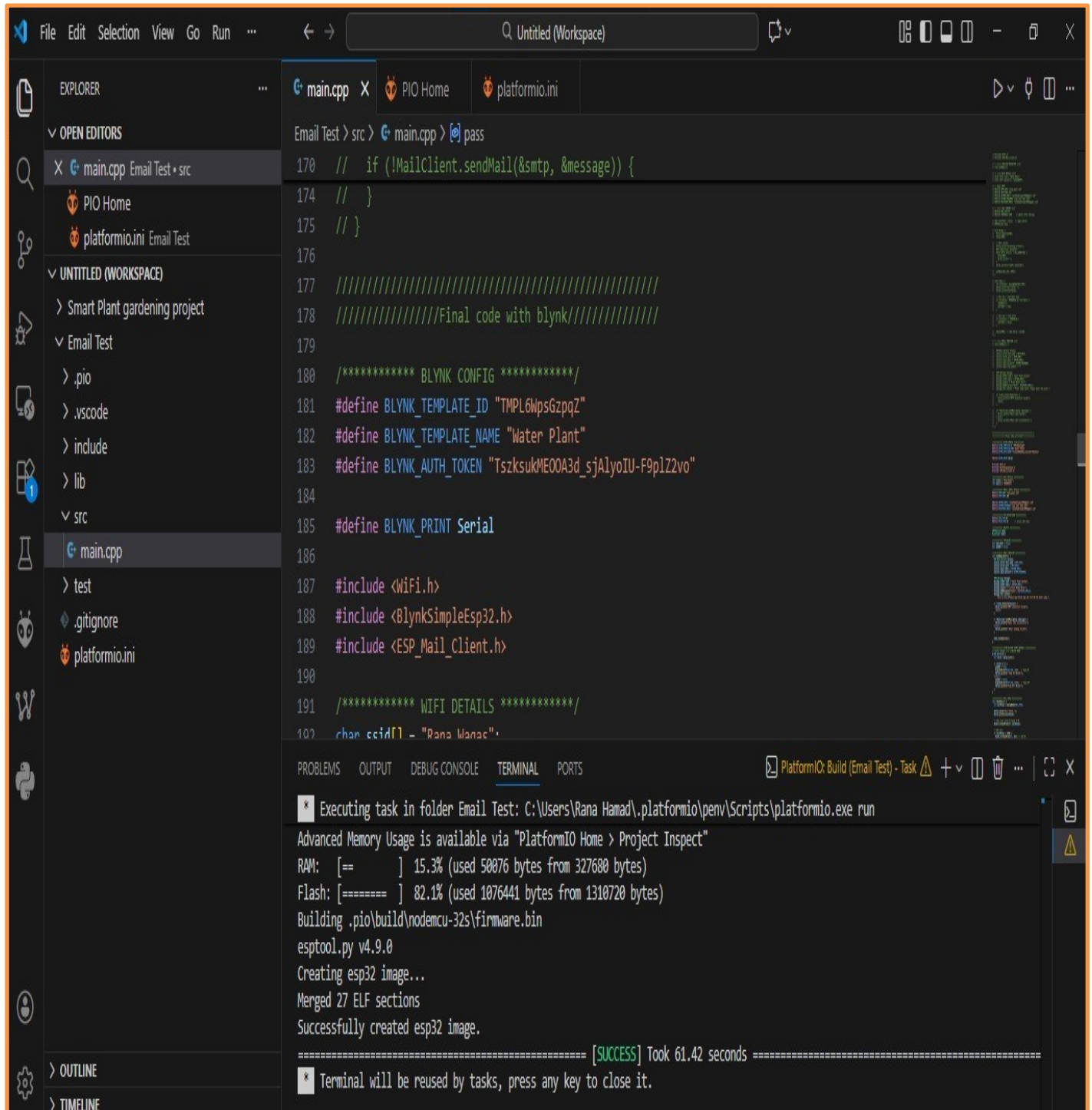


The image shows a Visual Studio Code editor window with a project for an ESP32. The Explorer sidebar on the left shows the file structure: `.pio`, `.vscode`, `include`, `lib`, `src` (containing `main.cpp`), `test`, `.gitignore`, and `platformio.ini`. The main editor displays the `main.cpp` file with the following code:

```
src > main.cpp > password
/ void setup() {
48
49 void loop() {
50 // Sirf WiFi status check karo har 10 seconds
51 if (WiFi.status() == WL_CONNECTED) {
52   Serial.print("WiFi Connected | IP: ");
53   Serial.println(WiFi.localIP());
54 } else {
55   Serial.println("❌ WiFi Disconnected!");
56 }
57
58 delay(10000); // 10 seconds wait
59 }
```

At the bottom, the TERMINAL panel shows the execution output:

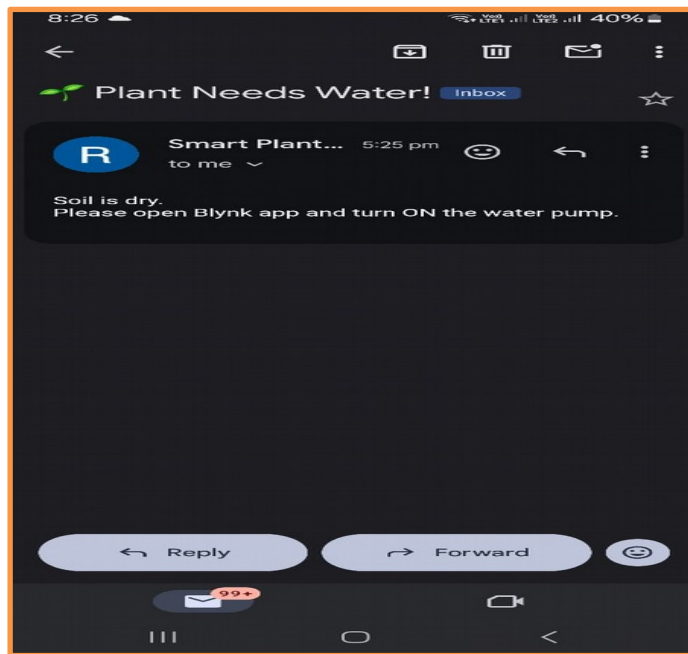
```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
* Executing task: C:\Users\Rana Hamad\.platformio\penv\Scripts\platformio.exe device monitor
ESP32 WIFI TEST PROGRAM
=====
Connecting to WiFi: Rana Waqas
.....
✅ WIFI CONNECTED SUCCESSFULLY!
IP Address: 192.168.100.149
Signal Strength (RSSI): -83 dBm
WiFi Connected | IP: 192.168.100.149
WiFi Connected | IP: 192.168.100.149
```



The serial monitor shows:

- Soil raw values
- Pump ON/OFF status
- WiFi connection status

5.3 Email Alert Output



with a message indicating watering is

6. Results, Conclusion, and Future Scope

6.1 Results

The system successfully:

- Monitored soil moisture accurately
- Sent email alerts when soil was dry
- Allowed remote pump control from any location
- Automatically turned OFF pump when soil became wet
- Reduced water wastage

6.2 Conclusion

The Smart Plant Gardening System provides an efficient and reliable solution for plant watering using IoT technology. It reduces manual effort and ensures proper plant care even when the user is not physically present. The integration of sensors, ESP32, cloud services, and mobile application makes the system smart and user-friendly.

6.3 Future Scope

- The system can be further improved by:
- Adding temperature and humidity sensors
- Integrating OLED display

- Using mobile push notifications
- Supporting multiple plants
- Adding solar power supply
- Implementing AI-based watering decisions